Optimization for Deep Learning (OPT4DL)

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Introduction to Stochastic Gradient Descent (SGD)

Motivation



- In ML/DL our loss function typically has the form

$$f(w) = \mathbb{E}_z[\ell(w, z)] \tag{1}$$

  (recall, think of it as average loss over data, i.e., $\frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i)$)
- To run GD, we need the gradient $\nabla \mathbb{E}_z[\ell(w, z)]$, which could be difficult to calculate
  - Large n means running out of memory and/or lots of calculations
  - In some cases we may not be able to calculate the expectation exactly (hard integrals)
- The main idea behind most training methods is to find good estimators such that $\hat{\nabla} \approx \nabla \mathbb{E}_z[\ell(w, z)]$



Proposition

If ℓ is a differentiable function of w and z, we can switch the order of derivative and expectation:

$$\nabla \mathbb{E}_z[\ell(w,z)] = \mathbb{E}_z[\nabla \ell(w,z)] \tag{2}$$

- This leads to leveraging sample approximation to estimate the expectation

$$z^1, \ldots, z^B \sim p_z, \qquad \mathbb{E}_z[\nabla \ell(w,z)] \approx \frac{1}{B} \sum_{j=1}^{B} \nabla \ell(w, z^j) \tag{3}$$

Stochastic Gradient Descent (SGD)



- Initialize w_1, learning rate $0 < \eta_t \leq 1$, and batch size $B \geq 1$
- For $t = 1, \ldots, T$ perform:
  - Sample a mini-batch of samples $z_t^1, \ldots, z_t^B \sim p_z$
  - Form the update vector $g_t = \frac{1}{B} \sum_{j=1}^{B} \nabla \ell(w_t, z_t^j)$
  - Update the parameters $w_{t+1} = w_t - \eta g_t$

Which Estimators are Good?



- We like estimators that have low bias (ideally zero) and low variance

Bias

Let X be a quantity and Y a Random Estimator of X. The bias of Y is defined as

$$\text{Bias}_Y = \|X - \mathbb{E}[Y]\| \tag{4}$$

Notably, Y is unbiased if $X = \mathbb{E}[Y]$.

Variance

Let X be a quantity and Y a Random Estimator of X. The variance of Y is defined as

$$\text{Var}(Y) = \mathbb{E}\|Y - \mathbb{E}[Y]\|^2 \tag{5}$$

Notably, if Y is unbiased, i.e., $X = \mathbb{E}[Y]$, we have $\text{Var}(Y) = \mathbb{E}\|Y - X\|^2$.

Which Estimators are Good? (Cont'd)



- We can characterize the total error of an estimators using bias-variance decomposition:

$$\text{Total error} = \mathbb{E}\|Y - X\|^2 = \|X - \mathbb{E}[Y]\|^2 + \mathbb{E}\|Y - \mathbb{E}[Y]\|^2 = \text{bias plus variance} \tag{6}$$

- Averaging independent unbiased estimators could be useful in lowering the error

$$\begin{aligned}
\mathbb{E}\|\frac{Y_1 + Y_2}{2} - X\|^2 = \mathbb{E}\|\frac{(Y_1 - X) + (Y_2 - X)}{2}\|^2 &= \frac{1}{4}\mathbb{E}\|(Y_1 - X) + (Y_2 - X)\|^2 \\
&= \frac{1}{4}\left(\mathbb{E}\|(Y_1 - X)\|^2 + \mathbb{E}\|(Y_2 - X)\|^2 + \mathbb{E}[\langle Y_1 - X, Y_2 - X\rangle]\right) \\
&= \frac{1}{4}\left(\mathbb{E}\|(Y_1 - X)\|^2 + \mathbb{E}\|(Y_2 - X)\|^2\right)
\end{aligned} \tag{7}$$

Why SGD is a Good Idea?



- Each estimator $Y_j = \nabla \ell(w_t, z_t^j)$ is an unbiased estimator of loss gradient if we sample from the distribution
- The average of Y_j's will also remain unbiased
- As $B \to \infty$, by strong law of large numbers our estimator $\frac{1}{N} \sum_{j=1}^{B} Y_j$ (almost surely) approaches the loss gradient.
- But B cannot be too large in practice when optimizing model parameters. Impact on training and test errors?



Stochastic First-Order Oracle (SFO)

Given the current parameter w_t and the sample/data used in iteration t, i.e., z_t, Oracle returns an update vector g_t which is conditionally unbiased with bounded variance:

$$\mathbb{E}_{z_t}[g_t|w_t] = \nabla f(w_t), \qquad \mathbb{E}_{z_t}[\|g_t - \nabla f(w_t)\|^2|w_t] \leq \sigma^2 \tag{8}$$

for some $0 \leq \sigma^2 < \infty$.

- SFO enables formal study of not just SGD, but also more advanced training methods
- A simple way to model gradient calculation in practice
- We need the conditioning since w_t is itself random: it depends on our (random) selection of mini batches in previous iterations. In other words, w_t is a function of $z_1, \ldots, z_{t-1}$ (note the exclusion of z_t).
- Thus, the following expectations are equivalent: $\mathbb{E}_{z_t}[\cdot|w_t] \equiv \mathbb{E}_{z_t}[\cdot|z_1, \ldots, z_{t-1}]$

Performance of SGD



Theorem

Assume f is G-Lipschitz and convex. Furthermore assume we have access to an SFO and that $z_1, \ldots, z_T$ are statistically independent. Then, SGD with the update rule $w_{t+1} = w_t - \eta g_t$ and learning rate $\eta = \frac{D}{\sqrt{T(G^2+\sigma^2)}}$ satisfies the following bound on the incurred average suboptimality

$$\mathbb{E}_{z_1,\ldots,z_T}[\frac{1}{T}\sum_{t=1}^{T} f(w_t) - f(w^*)] \leq \frac{D\sqrt{G^2+\sigma^2}}{\sqrt{T}} \leq \frac{DG}{\sqrt{T}} + \frac{D\sigma}{\sqrt{T}} \tag{9}$$

- As our method is a randomized algorithm, our statement regarding its performance needs to be stated either on expectation or with high confidence
- As we discussed, with larger batch sizes $\sigma^2 \to \sigma^2/B$
- We can use either of 3 approaches to generate an output, that is (i) the last/best parameter w_T, (ii) averaging all parameters $w_1, \ldots, w_T$, and (iii) sampling one iterate at random

Minibatch SGD: A Fixed Budget Analysis



- Recall after T iterations of mini batch SGD

$$\text{error} \leq \frac{DG}{\sqrt{T}} + \frac{D\sigma}{\sqrt{BT}} \tag{10}$$

- What is the best B if we fix the total computation budget, i.e., number of iterations times number of stochastic gradient calculation per iteration: $C = B \times T$ (also called number of SFO calls)
- In terms of C, the error becomes

$$\text{error} \leq \frac{DG\sqrt{B}}{\sqrt{C}} + \frac{D\sigma}{\sqrt{C}} \tag{11}$$

  Suggesting the best batch size is $B = 1$?
- Why mini batch SGD with $B > 1$ then?

Answer: Parallel computation



- Assume we have M machines and that the computing time scales linearly with M

$$\tau = \frac{C}{M} \tag{12}$$

- Assume $M = B$ our batch size. Then, in terms of the computing time, the error becomes

$$\text{error} \leq \frac{DG}{\sqrt{\tau}} + \frac{D\sigma}{\sqrt{\tau B}} \tag{13}$$

- Larger batch sizes plus parallel computation reduces the error faster
- Remark: an analysis based on worst case scenario
- $B > 1$ leads to lower variance and smoother convergence (less ups and downs)

SGD in Practice (Reshuffling)



Consider a training problem, i.e. ERM:

$$\min_w f(w) = \frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i) \tag{14}$$

- For $e = 1, \ldots, E$ epochs do
  - For $t = 1, \ldots, K$ inner iterations do
    1. form the mini-batch of samples $z_t^1, \ldots, z_t^B \sim p_z$
    2. Form the update vector $g_t = \frac{1}{B}\sum_{j=1}^{B} \nabla \ell(w_t, z_t^j)$
    3. Update the parameters $w_{t+1} = w_t - \eta g_t$
  - Optionally, shuffle the data into a new random order)

Has been observed to lead to smoother loss decay and better generalization

This can be attributed to the lower variance of the resultant (and more complex) gradient estimator, due to reshuffling

- We will leverage our general proposition:

$$\sum_{t=1}^{T}\langle g_t, w_t - w^*\rangle \leq \frac{1}{2\eta}\|w_1 - w^*\|^2 + \frac{\eta}{2}\sum_{t=1}^{T}\|g_t\|^2 \tag{15}$$

  with suitable conditional expectations to account for the randomization of SGD.

- Let $z_{1:t}$ denote $z_1, \ldots, z_t$. We have by tower expectation, linearity of expectation, and the unbiasedness property of the SFO

$$\begin{aligned}\mathbb{E}_{z_{\mathbf{1}:t}}[\langle g_t, w_t - w^*\rangle] &= \mathbb{E}_{z_{\mathbf{1}:t-\mathbf{1}}}[\mathbb{E}_{z_t}[\langle g_t, w_t - w^*\rangle | z_{1:t-1}]] = \mathbb{E}_{z_{\mathbf{1}:t-\mathbf{1}}}[\langle \mathbb{E}_{z_t}[g_t | z_{1:t-1}], w_t - w^*\rangle] \\ &= \mathbb{E}_{z_{\mathbf{1}:t-\mathbf{1}}}[\langle \nabla f(w_t), w_t - w^*\rangle]\end{aligned} \tag{16}$$

Proof of Theorem (Cont'd)

- Recall, by convexity

$$f(w_t) - f(w^*) \leq \langle \nabla f(w_t), w_t - w^* \rangle \tag{17}$$

- Thus

$$\mathbb{E}_{z_{1:t-1}}[f(w_t) - f(w^*)] \leq \mathbb{E}_{z_{1:t-1}}[\langle \nabla f(w_t), w_t - w^* \rangle] \tag{18}$$

- Using the independence of samples used in each iteration (w_t only a function of $z_{1:t-1}$ and independent from the future samples $z_t, \ldots, z_T$)

$$\mathbb{E}_{z_1,\ldots,z_T}[f(w_t) - f(w^*)] = \mathbb{E}_{z_{1:t-1}}[f(w_t) - f(w^*)] \tag{19}$$

- Thus, by linearity of expectation and our general proposition

$$\mathbb{E}_{z_1,\ldots,z_T}[\sum_{t=1}^{T} f(w_t) - f(w^*)] \leq \frac{1}{2\eta}\|w_1 - w^*\|^2 + \frac{\eta}{2}\sum_{t=1}^{T}\mathbb{E}_{z_1,\ldots,z_T}\|g_t\|^2 \tag{20}$$

Proof of Theorem (Cont'd)



- For the second term note by the independence of the samples (g_t only a function of w_t and z_t)

$$\mathbb{E}_{z_1,\ldots,z_T}\|g_t\|^2 = \mathbb{E}_{z_1,\ldots,z_t}\|g_t\|^2 \tag{21}$$

- By tower expectation

$$\mathbb{E}_{z_1,\ldots,z_t}\|g_t\|^2 = \mathbb{E}_{z_1,\ldots,z_{t-1}}[\mathbb{E}_{z_t}[\|g_t\|^2|z_1,\ldots,z_{t-1}]] \tag{22}$$

- We will now aim to use our unbiased SFO assumption with bounded variance,

$$\begin{aligned}\mathbb{E}_{z_t}[\|g_t\|^2|z_1,\ldots,z_{t-1}] &= \mathbb{E}_{z_t}[\|g_t \pm \nabla f(w_t)\|^2|z_1,\ldots,z_{t-1}] \\ &= \mathbb{E}_{z_t}[\|\nabla f(w_t)\|^2|z_1,\ldots,z_{t-1}] + \mathbb{E}_{z_t}[\|g_t - \nabla f(w_t)\|^2|z_1,\ldots,z_{t-1}] \\ &\leq G^2 + \sigma^2\end{aligned} \tag{23}$$

by Lipschitzness and bounded variance assumptions.